

Assignment 3 — C Programming Error Analysis

Problems 1–10 | Syntax · Semantic · Logical · Runtime Errors

Q.No.	Line No.	Error	Error Type	Error Correction
1	8	Missing semicolon after 'int x = 10'	Syntax	int x = 10;
1	13	Assignment used instead of comparison (x = 10)	Semantic	if(x == 10)
1	17	Assigning string literal to int variable	Semantic	int marks = 90;
1	19	Using undeclared variable 'z'	Semantic	Declare 'int z;' before use
1	21	Assignment to constant (100 = x)	Syntax	x = 100;
1	25	Array index out of bounds (arr[10] for arr[5])	Runtime	Use index 0–4 only, e.g. arr[4] = 50;
1	27	Assigning float pointer to int variable without cast	Semantic	float *ptr = (float *)&x; or use correct type
1	29	Using %f format specifier for int variable x	Semantic	printf("%d\n", x);
1	33	Division by zero (a/b where b=0)	Runtime	Ensure b != 0 before dividing
1	35	Multi-character character constant 'AB'	Syntax	char ch = 'A';
1	39	Logical error: both if and else print 'X is greater'	Logical	else printf("Y is greater");
1	42	Return value of sqrt() not used / missing printf	Logical	printf("%f\n", sqrt(25));
1	44	Calling undeclared function 'show()'	Semantic	Declare/define show() before calling
1	46	Semicolon after while condition causes infinite loop	Logical	Remove ';': while(x < 15) {
1	49	'continue' used outside a loop	Syntax	Remove the continue; statement
1	51	'break' used outside a loop/switch	Syntax	Remove the break; statement
1	53	Function called with 1 argument but requires 2 (calculate)	Semantic	int result = calculate(10, 20);
2	9	Missing semicolon after 'int num2 = 20'	Syntax	int num2 = 20;
2	21	Missing closing ')' in if(choice == 1	Syntax	if(choice == 1)
2	31	Extra closing ')' in else if(choice == 3))	Syntax	else if(choice == 3)
2	43	Missing closing ')' in for loop: for(int i=0; i<5; i++	Syntax	for(int i=0; i<5; i++)
2	49	Missing closing ')' in while(choice > 0	Syntax	while(choice > 0)
2	54	Missing semicolon after while(choice > 5) in do-while	Syntax	while(choice > 5);

Q.No.	Line No.	Error	Error Type	Error Correction
2	62	Missing colon after 'case 2'	Syntax	case 2:
2	77	Unclosed character literal char grade = 'A;	Syntax	char grade = 'A';
2	80	Missing ']' in array declaration int arr[5;	Syntax	int arr[5];
2	91	Missing closing '}' for inner block (unmatched brace)	Syntax	Add '}' to close the block
2	99	Missing semicolon after last printf	Syntax	printf("End of Program\n");
3	8	Assigning string literal to int variable 'age'	Semantic	int age = 20;
3	12	Assigning string literal to char variable 'grade'	Semantic	char grade = 'A';
3	14	Using %d format for float variable 'salary'	Semantic	printf("%f\n", salary);
3	16	Using %f format for int variable 'marks'	Semantic	printf("%d\n", marks);
3	20	Using undeclared variable 'score'	Semantic	Declare 'int score;' before use
3	23	Re-declaration of variable 'x'	Semantic	Remove the second 'int x = 20;'
3	25	Assignment to constant (100 = x)	Syntax	x = 100;
3	27	Using %s format for int variable 'num'	Semantic	printf("%d\n", num);
3	31	Using %d format for float variable 'pi'	Semantic	printf("%f\n", pi);
3	34	Using float as array index (arr[2.5])	Semantic	arr[2] = 100;
3	36	Assigning float 65.5 to char (loss of data)	Semantic	char ch = 'A'; // ASCII 65
3	39	Calling add() with 1 argument instead of 2	Semantic	result = add(10, 5);
3	44	Calling average() with 2 arguments instead of 3	Semantic	avg = average(10, 20, 30);
3	46	Assigning string literal to int variable 'temperature'	Semantic	int temperature = 35;
3	49	Assigning string literal to float variable 'discount'	Semantic	float discount = 10.5;
3	52	Direct string assignment to char array (name = "Robil")	Semantic	strcpy(name, "Robil");
3	56	Using undeclared variable 'totalMarks'	Semantic	Declare 'int totalMarks;' before use
3	59	Using %c format for int variable 'number'	Semantic	printf("%d\n", number);
3	62	Using %d format for float variable 'rate'	Semantic	printf("%f\n", rate);
3	66	Calling add(a,b,c) with 3 args but function takes only 2	Semantic	sum = add(a, b);
3	73	Assigning float result to int 'area' (precision loss)	Semantic	float area = 3.14 * radius * radius;

Q.No.	Line No.	Error	Error Type	Error Correction
3	78	Assignment used instead of comparison (studentMarks = 100)	Semantic	if(studentMarks == 100)
3	82	Assigning string literal to int variable 'size'	Semantic	int size = 5;
3	85	Using %d format for float variable 'percentage'	Semantic	printf("%f\n", percentage);
3	88	Using string as array index (arr2["five"])	Semantic	arr2[5] = 100;
3	95	Using string in case label (case "2":)	Syntax	case 2:
3	100	Calling undeclared function 'multiply()'	Semantic	Declare/define multiply() before calling
3	106	Using %d format for float variable 'areaRect'	Semantic	printf("%f\n", areaRect);
3	113	Assigning float return to int 'avgMarks' from average()	Semantic	float avgMarks = average(marks1, marks2, marks3);
3	116	Using undeclared variable 'value'	Semantic	Declare 'int value;' before use
3	119	Assigning string literal to int variable 'employeeId'	Semantic	int employeeId = 101;
4	35	Calling undeclared/undefined function 'generateReport()'	Semantic	Declare and define generateReport() function
4	36	Calling undeclared/undefined function 'displayEmployee()'	Semantic	Declare and define displayEmployee() function
4	37	Calling undeclared/undefined function 'processData()'	Semantic	Declare and define processData() function
4	38	Calling undeclared/undefined function 'saveRecord()'	Semantic	Declare and define saveRecord() function
4	39	Calling undeclared/undefined function 'loadRecord()'	Semantic	Declare and define loadRecord() function
4	40	Calling undeclared/undefined function 'printInvoice()'	Semantic	Declare and define printInvoice() function
4	41	Calling undeclared/undefined function 'calculateTax()'	Semantic	Declare and define calculateTax() function
4	42	Calling undeclared/undefined function 'validateUser()'	Semantic	Declare and define validateUser() function
4	43	Calling undeclared/undefined function 'sendNotification()'	Semantic	Declare and define sendNotification() function
4	44	Calling undeclared/undefined function 'generateCertificate()'	Semantic	Declare and define generateCertificate() function
4	55	Missing definition for declared function 'calculateSalary()'	Semantic	Define calculateSalary(float basic) { ... }
4	55	Missing definition for declared function 'printStudentDetails()'	Semantic	Define printStudentDetails() { ... }
4	55	Missing definition for declared function 'showResult()'	Semantic	Define showResult() { ... }
4	55	Missing definition for declared function 'calculateAverage()'	Semantic	Define calculateAverage(int a, int b, int c) { ... }
4	13	extern variable 'companyProfit' declared but never defined	Semantic	Add definition: float companyProfit = 0.0; somewhere
5	12	Division by zero at runtime (a/b where b=0)	Runtime	Check b != 0 before dividing

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5	16	Dereferencing NULL pointer ptr1	Runtime	Allocate memory: <code>int *ptr1 = malloc(sizeof(int));</code>
5	20	Array out of bounds: <code>arr[10]</code> for <code>arr[5]</code>	Runtime	Use valid index 0–4
5	23	Dereferencing uninitialized pointer ptr2	Runtime	Initialize: <code>int *ptr2 = malloc(sizeof(int));</code>
5	27	Buffer overflow: copying 'Programming' into <code>char[5]</code>	Runtime	<code>char name[12];</code> or copy shorter string
5	31	Reading uninitialized pointer ptr3	Runtime	Initialize ptr3 before dereferencing
5	37	Using NULL file pointer fp without NULL check	Runtime	<code>if(fp == NULL) { perror("Error"); return 1; }</code>
5	43	Buffer overflow: copying 'Computer Science'(16) into <code>malloc(5)</code>	Runtime	<code>ptr4 = malloc(17);</code> // allocate enough space
5	48	Using freed pointer ptr4 (use-after-free)	Runtime	Do not access ptr4 after <code>free()</code>
5	53	Double free of ptr5	Runtime	Remove second <code>free(ptr5);</code>
5	58	Modulo by zero (<code>x % y</code> where <code>y=0</code>)	Runtime	Ensure <code>y != 0</code> before using <code>%</code>
5	62	Buffer overflow: <code>strcat</code> appends 'WorldProgramming' to <code>str1[10]</code>	Runtime	Increase str1 size: <code>char str1[30] = "Hello";</code>
5	68	Heap overflow: writing 10 ints into <code>malloc(3 * sizeof(int))</code>	Runtime	<code>malloc(10 * sizeof(int))</code> or loop to <code>i<3</code>
5	72	Matrix out of bounds: <code>matrix[5][5]</code> on <code>matrix[3][3]</code>	Runtime	Use valid indices 0–2 only
5	75	Dereferencing NULL pointer ptr6	Runtime	Allocate memory before use
5	79	<code>gets()</code> is unsafe — can cause buffer overflow	Runtime	Use <code>fgets(password, 8, stdin);</code> instead
5	84	Use-after-free: writing to freed ptr7	Runtime	Do not write to ptr7 after <code>free()</code>
5	87	Reading uninitialized variable 'age'	Runtime	Initialize: <code>int age = 0;</code>
5	90	Negative array index <code>arr[-1]</code>	Runtime	Use non-negative index only
5	93	Dereferencing uninitialized pointer ptr8	Runtime	Initialize ptr8 before use
5	96	Printing NULL char pointer with <code>%s</code>	Runtime	Initialize: <code>char *text = "some text";</code>
5	100	Possible NULL return from <code>malloc</code> not checked	Runtime	Check <code>hugeArray != NULL</code> after <code>malloc</code>
5	105	Infinite loop incrementing NULL pointer ptr9	Runtime	Remove or fix the infinite loop
6	12	Wrong condition: <code>num%2==1</code> means odd, not even	Logical	<code>if(num % 2 == 0) printf("Number is Even");</code>
6	22	Grade logic inverted: <code>marks>=90</code> prints Grade B (should be A)	Logical	<code>if(marks >= 90) printf("Grade A\n");</code> // etc.
6	34	Leap year check incomplete (divisible by 100/400 missing)	Logical	<code>if((year%4==0 && year%100!=0) (year%400==0))</code>
6	42	Voting eligibility: <code>age>18</code> prints Not Eligible (should be Eligible)	Logical	<code>if(age >= 18) printf("Eligible for Voting\n");</code>

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6	50	Smallest number logic is inverted (uses a>b to find smaller)	Logical	if(a < b) printf("Smaller = %d", a); else printf("Smaller = %d", b);
6	60	Largest logic wrong: uses < to find largest (should use >)	Logical	if(y > largest) largest = y; if(z > largest) largest = z;
6	68	Integer division loses decimal for average	Logical	average = (float)totalMarks / subjects;
6	74	Circle area formula wrong: should be πr^2 not $2\pi r^2$	Logical	area = 3.14 * radius * radius;
6	81	Bonus condition logic inverted (salary>30000 gives 2% instead of 20%)	Logical	if(salary <= 30000) bonus = salary*0.20; else bonus = salary*0.02;
6	90	Switch month cases off by one (case 1 prints February etc.)	Logical	case 1: printf("January"); case 2: printf("February"); etc.
6	103	Prime check sets isPrime=1 when divisor found (should be 0)	Logical	isPrime = 0; break; // inside if(n%i==0)
6	109	Discount is added instead of subtracted	Logical	finalAmount = purchaseAmount - discount;
6	117	Sum loop assigns i to sum instead of adding (sum = i)	Logical	sum += i;
6	125	Temperature logic inverted: <40 prints Cold (35 would be Hot)	Logical	if(temperature >= 40) printf("Hot Weather"); else printf("Cold Weather");
6	131	Grade 'A' message inverted: prints 'Average' for A	Logical	if(grade == 'A') printf("Excellent Performance\n");
6	137	Rectangle area uses + instead of *	Logical	rectangleArea = length * width;
6	143	Pass/Fail condition inverted: >=60 prints Fail	Logical	if(percentage >= 60) printf("Pass\n"); else printf("Fail\n");
7	8	Array out of bounds in loop: i<=5 accesses marks[5]	Runtime	for(int i=0; i<5; i++)
7	18	Integer division for average (total/5 loses decimal)	Logical	average = (float)total / 5;
7	27	Loop index starts at 1 causing numbers[0] uninitialized and numbers[10] out of bounds	Runtime	for(int i=0; i<10; i++) numbers[i] = (i+1)*10;
7	33	Buffer overflow: 'ComputerScience'(15) into char name[10]	Runtime	char name[16]; or use shorter string
7	40	Direct string assignment to char array (city = "Delhi")	Semantic	strcpy(city, "Delhi");
7	47	Comparing strings with == instead of strcmp()	Logical	if(strcmp(str1, str2) == 0)
7	53	gets() is unsafe — can cause buffer overflow	Runtime	fgets(password, 8, stdin);
7	63	grade[] not null-terminated, printf("%s") may print garbage	Runtime	grade[2] = '\0'; // null-terminate
7	69	Negative array index arr[-1] — undefined behavior	Runtime	Use valid index >= 0
7	76	Buffer overflow: 'student@gmail.com'(17) into email[15]	Runtime	char email[20] = "student@gmail.com";
7	82	scores[3] out of bounds for scores[3] (valid: 0–2)	Runtime	Remove scores[3]=60; and access only 0–2

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7	86	Buffer overflow: 'Programming'(11) into subject[10]	Runtime	char subject[12] = "Programming";
7	92	Comparing string with == instead of strcmp()	Logical	if(strcmp(username, "admin") == 0)
7	98	Buffer overflow: '987654321012'(12) into mobile[10]	Runtime	char mobile[13]; or use shorter number
7	104	Off-by-one: loop i<=5 accesses attendance[5] out of bounds	Runtime	for(int i=0; i<5; i++)
7	110	Out-of-bounds access: course[20] on char[20] (valid 0–19)	Runtime	printf("%c\n", course[19]); or course[0]
7	116	college[] not null-terminated — printf("%s") undefined	Runtime	college[5] = '\0';
7	126	section[3] too small for 'IT-A'(4 chars + null)	Runtime	char section[5] = "IT-A";
8	15	Calling add() with 1 argument instead of 2	Semantic	result = add(num1, num2);
8	20	Calling average() with 2 arguments instead of 3	Semantic	avg = average(70, 80, 90);
8	33	swap(x,y) passes by value — swap has no effect	Logical	Change swap to take pointers: swap(&x, &y); and update definition
8	41	Calling undeclared function 'calculateGrade()'	Semantic	Declare/define calculateGrade() before calling
8	45	Calling undeclared function 'circleArea()'	Semantic	Declare/define circleArea() before calling
8	50	Calling undeclared function 'sumArray()'	Semantic	Declare/define sumArray() before calling
8	55	Calling undeclared function 'displayResult()'	Semantic	Declare/define displayResult() before calling
8	64	factorial(0) returns 0 instead of 1	Logical	if(n == 0) return 1;
8	78	maximum() returns wrong value (returns smaller number)	Logical	if(a > b) return a; else return b;
8	87	fibonacci() infinite recursion: fibonacci(n)+fibonacci(n-1) should be (n-1)+(n-2)	Logical	return fibonacci(n-1) + fibonacci(n-2);
8	96	simpleInterest() returns nothing (missing return value)	Logical	return si;
8	107	power() loop starts at i=1, calculates base^(exp-1)	Logical	for(int i=0; i<exp; i++) // start from 0
8	119	reverseNumber() has no return statement	Logical	return rev; // at end of function
8	139	isPrime() returns 1 when divisor found (should return 0)	Logical	return 0; // not prime when divisible
8	147	updateValue() passes by value — change has no effect	Logical	void updateValue(int *x) { *x = 500; }
8	157	printArray() loop i<=size causes out-of-bounds access	Runtime	for(int i=0; i<size; i++)
8	162	recursiveDemo() has infinite recursion (no base case)	Runtime	Add base case: if(n <= 0) return 0;
8	170	rectangleArea() returns sum instead of product	Logical	return length * width;

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8	175	cube() returns n*n (square) instead of n*n*n	Logical	return n * n * n;
9	12	Reading uninitialized pointer ptr1 — undefined behavior	Runtime	Initialize: int val = 0; int *ptr1 = &val;
9	16	Dereferencing NULL pointer ptr2	Runtime	ptr2 = malloc(sizeof(int)); before *ptr2 = 100;
9	22	Incrementing ptr3 past x — may access garbage memory	Runtime	Do not increment ptr3 past known memory
9	27	Out-of-bounds: *(ptr4+10) on array of size 5	Runtime	Access only indices 0–4
9	34	Use-after-free: reading *ptr5 after free(ptr5)	Runtime	Do not access ptr5 after free()
9	39	Memory leak: ptr6 set to NULL without freeing allocated memory	Runtime	free(ptr6); before ptr6 = NULL;
9	43	Heap overflow: writing 10 ints into malloc(5 bytes)	Runtime	ptr7 = malloc(10 * sizeof(int));
9	49	Out-of-bounds: ptr8[3] on 3-element array (valid: 0–2)	Runtime	Remove ptr8[3] = 40;
9	55	Double free of ptr9	Runtime	Remove the second free(ptr9);
9	59	Reading uninitialized heap memory from ptr10	Runtime	Initialize after malloc: *ptr10 = 0;
9	64	Off-by-one: i<=5 writes numbers[5] out of bounds	Runtime	for(int i=0; i<5; i++)
9	72	pptr assigned address of int, but pptr is int** — type mismatch	Semantic	int *valptr = &value; int **pptr = &valptr;
9	77	Buffer overflow: strcpy 'Programming'(11) into malloc(5)	Runtime	name = malloc(12);
9	82	Incrementing ptr11 after malloc — lost track of original pointer	Runtime	Do not modify ptr11 after malloc; use index instead
9	86	Dereferencing NULL pointer ptr12 in if condition	Runtime	Check: if(ptr12 != NULL && *ptr12 > 0)
9	92	Out-of-bounds: ptr13[100] on 100-element array (valid: 0–99)	Runtime	Remove ptr13[100] = 500;
9	97	Use-after-free: writing to ptr14 after free(ptr14)	Runtime	Do not write to ptr14 after free()
9	101	p assigned matrix (int(*)[3]) but p is int* — type mismatch	Semantic	int *p = &matrix[0][0];
9	106	Out-of-bounds: *(p+20) on 3x3 matrix (9 elements total)	Runtime	Access only indices 0–8
9	110	Memory leak: ptr15 first malloc overwritten without freeing	Runtime	free(ptr15); before second malloc
9	116	swap() swaps local pointers, not the values pointed to	Logical	Use temp value: int temp=*a; *a=*b; *b=temp;
9	121	printArray() loop i<=size goes out of bounds	Runtime	for(int i=0; i<size; i++)
10	12	Missing semicolon after 'int choice = 1'	Syntax	int choice = 1;
10	14	Assigning string literal to int variable 'marks'	Semantic	int marks = 90;
10	20	Calling add() with 1 argument instead of 2	Semantic	result = add(a, b);

Q.No.	Line No.	Error	Error Type	Error Correction
10	24	Assignment instead of comparison (a = b)	Semantic	if(a == b)
10	28	Using undeclared variable 'score'	Semantic	Declare 'int score;' before use
10	32	Array out of bounds: arr[10] for arr[5]	Runtime	Use index 0–4 only
10	34	Negative array index arr[-1] — undefined behavior	Runtime	Use valid non-negative index
10	38	Buffer overflow: 'ProgrammingLanguage'(19) into name[10]	Runtime	char name[20]; or use shorter string
10	43	Direct string assignment to char array (city = "Delhi")	Semantic	strcpy(city, "Delhi");
10	49	Comparing string with == instead of strcmp()	Logical	if(strcmp(user, "admin") == 0)
10	54	Division by zero (x/y where y=0)	Runtime	Ensure y != 0 before dividing
10	57	Dereferencing NULL pointer ptr1	Runtime	ptr1 = malloc(sizeof(int)); before *ptr1 = 500;
10	61	Reading uninitialized pointer ptr2	Runtime	Initialize: int val; int *ptr2 = &val;
10	66	Use-after-free: reading *ptr3 after free(ptr3)	Runtime	Do not access ptr3 after free()
10	68	Double free of ptr3	Runtime	Remove the second free(ptr3);
10	72	Heap overflow: writing 10 ints into malloc(5 bytes)	Runtime	ptr4 = malloc(10 * sizeof(int));
10	77	Memory leak: ptr5 first malloc overwritten without freeing	Runtime	free(ptr5); before second malloc call
10	81	Calling average() with 2 args instead of 3	Semantic	avg = average(70, 80, 90);
10	86	Voting eligibility inverted: age>18 prints Not Eligible	Logical	if(age >= 18) printf("Eligible\n"); else printf("Not Eligible\n");
10	93	Prime check sets isPrime=1 when factor found (should be 0)	Logical	isPrime = 0; break;
10	97	isPrime message inverted: prints 'Not Prime' when prime	Logical	if(isPrime) printf("Prime\n"); else printf("Not Prime\n");
10	101	Calling undeclared function 'calculateSalary()'	Semantic	Declare/define calculateSalary() before use
10	108	Missing colon after 'case 1' in switch	Syntax	case 1:
10	114	case 2 has no break — falls through to default	Logical	Add break; after printf("Two\n");
10	118	Multi-character char literal 'AB' assigned to char grade	Syntax	char grade = 'A';
10	122	Out-of-bounds: scores[3]=40 for scores[3] (valid: 0–2)	Runtime	Remove scores[3]=40; or resize to scores[4]
10	127	Off-by-one in loop: i<=3 accesses scores[3] out of bounds	Runtime	for(int i=0; i<3; i++)
10	133	Circle area formula uses 2*PI*r^2 (sphere) instead of PI*r^2	Logical	area = 3.14 * radius * radius;
10	137	Semicolon after while condition causes infinite loop	Logical	Remove ';': while(choice > 0) {

Q.No.	Line No.	Error	Error Type	Error Correction
10	140	'continue' used outside a loop	Syntax	Remove the continue; statement